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Original Contribution

Deep Learning Analysis of Prenatal Ultrasound for Identification of Ventriculomegaly

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ABSTRACT

Objective: To develop and evaluate a deep learning model capable of detecting ventriculomegaly on prenatal ultrasound images using a foundation model pre-trained specifically on ultrasound data.

Methods: A vision transformer-based ultrasound self-supervised foundation model with masked autoencoding (USF-MAE) was fine-tuned for binary classification of fetal brain ultrasound images as normal or ventriculomegaly. The encoder had been pre-trained on more than 370,000 ultrasound images from the OpenUS-46 corpus. For this study, it was adapted and fine-tuned on a curated data set of fetal brain images. Model performance was evaluated through fivefold cross-validation as well as an independent test cohort. Accuracy, precision, recall, specificity, F_1 -score and area under the receiver operating characteristic and precision-recall curves were reported as the performance metrics. Eigen-CAM and Grad-CAM were used to visualize model attention.

Results: USF-MAE reported an F_1 -score of 91.76% for the cross-validation set and 91.78% for the test set. The model outperformed all the baseline models, which included VGG-19, ResNet-50, ViT-B/16 and MoCo v3. The model reported a mean test precision of 94.47% and an accuracy of 97.24%. Activation maps indicated that the model consistently focused on the ventricular region when identifying ventriculomegaly.

Conclusion: Pre-training on a large, ultrasound-specific corpus improved classification performance and generalization for ventriculomegaly detection. The USF-MAE framework provided strong accuracy, reliability and explainability, showing potential as a robust tool for the assessment of fetal brain structures on prenatal ultrasound.

Introduction

Deep learning (DL) is a type of machine learning that uses artificial neural networks to automatically learn hierarchical feature representations from data, enabling a wide range of tasks such as prediction, classification, generation and reconstruction. Due to its exceptional pattern recognition and its ability to process large amounts of data, DL is increasingly used in medicine for medical image processing. For example, DL is used for the classification of images (e.g., normal or abnormal) to support diagnosis [1–5], object detection to annotate structures

(e.g., tumors) [6–8] and image segmentation to precisely contour structures [6,9,10–13]. It has already shown great success in diagnosing ophthalmologic conditions, breast cancer and lung nodules [14]. Convolutional neural networks (CNNs) and vision transformers (ViTs) are the main DL architectures used for such applications [15].

The general success of DL in medical image analysis has increased interest in its application to obstetric ultrasonography, and it offers an exciting opportunity to enhance prenatal diagnosis. However, unlike other imaging modalities such as computed tomography and magnetic resonance imaging, ultrasonography is very operator-dependent and

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thus has high inter-operator variability [16]. This introduces additional noise to the image data set, which might affect the accuracy and generalizability of the models [17,18]. Although several studies have explored the usefulness and performance of DL on fetal ultrasonography, continued investigation is needed to establish its clinical value.

Ventriculomegaly (VM) is defined as an enlargement of the lateral ventricles greater than 10 mm in the developing brain, and is associated with central nervous system and non-central nervous system malformations [19]. Identification of this condition is first done on ultrasound, the first-line imaging modality used during prenatal screening. VM is also among the diagnoses with the highest false-positive rate, which likely causes anxiety in parents, and may lead to unnecessary and potentially risky subsequent tests, highlighting the pressing need to improve the diagnostic accuracy of VM on ultrasound [19]. Other studies using DL for VM detection are either on computed tomography or magnetic resonance imaging [20], trained on relatively small data sets [21], measure lateral ventricles without classification [21] or are not specific to VM [22].

Our research team has successfully applied DL to the ultrasound diagnosis of cystic hygroma and congenital anomalies of the kidneys and urinary tract [23,24]. Recently, we developed an ultrasound self-supervised foundation model with masked autoencoding (USF-MAE), the first large-scale, MAE foundation model pre-trained exclusively on ultrasound data [25], that demonstrates strong cross-anatomical generalization and label efficiency across multiple ultrasound classification tasks. Building upon this foundation, the present study applied USF-MAE to the prenatal ultrasound diagnosis of VM. Our contributions are twofold:

- i) To extend the investigation of foundation model-based DL approaches for ultrasound image analysis by leveraging a pre-trained, self-supervised architecture specifically tailored to the ultrasound domain.
- ii) To fine-tune this USF-MAE for the binary classification of fetal brain ultrasound images as either normal or showing VM.

The performance of USF-MAE was further compared against common DL foundation models pre-trained on natural images, such as convolutional and transformer-based architectures, which lack domain-specific ultrasound feature representations. This comparison highlights the benefits of modality-specific pre-training for achieving higher accuracy and generalization in fetal neuroimaging tasks.

Methodology

Study setting and data acquisition

The study was approved by the Ottawa Health Science Network Research Ethics Board (OHSNREB #20210079). This retrospective study included ultrasound images from pregnant individuals with singleton or twin pregnancies obtained between June 2014 and May 2021 at a tertiary hospital in Eastern Ontario, Canada (the use of data remained within the hospital). All images were acquired using the GE HealthCare Voluson V730, E08 and E10 ultrasound systems with a transabdominal probe (C2-9) by professional obstetric sonographers following International Society of Ultrasound in Obstetrics and Gynecology guidelines for second-trimester fetal morphology assessment and fetal neurosonography, and confirmed by maternal–fetal medicine specialists. Among these, we selected fetal brain images with or without VM that were obtained between 18 and 24 wk of gestation (Fig. 1 shows an example of each class). Multiple images were collected from patients who underwent several ultrasound examinations within the designated gestational age range.

Images of a 2-D transventricular (axial) plane at the level displaying the frontal horns and cavum septi pellucidi, ensuring a symmetrical appearance of the cerebral hemispheres, were extracted from the institutional Picture Archiving and Communication System and saved in digital imaging and communication in medicine (DICOM) format. VM cases were identified as having an atrial diameter of ≥ 10 mm on the transventricular plane [27]. The images were classified into two groups, with or without VM, with no further classification based on severity. Classification was not based solely on qualitative assessment. Instead, it followed standardized biometric measurements and structured anatomical evaluation. Normal examinations required appropriate visualization of the standard intracranial planes (transventricular, transthalamic and trans-cerebellar), a normal cavum septi pellucidi, normal posterior fossa structures and a ventriculoatrial width < 10 mm. We also excluded ultrasound images of fetuses with brain anomalies other than VM, as well as those not acquired in standard grayscale ultrasound format.

Data set distribution and splitting strategy

The curated VM data set contained 823 fetal brain ultrasound images, of which 680 were normal and 143 represented cases of VM (Table 1). This corresponded to about a 1 : 5 class imbalance ratio, with

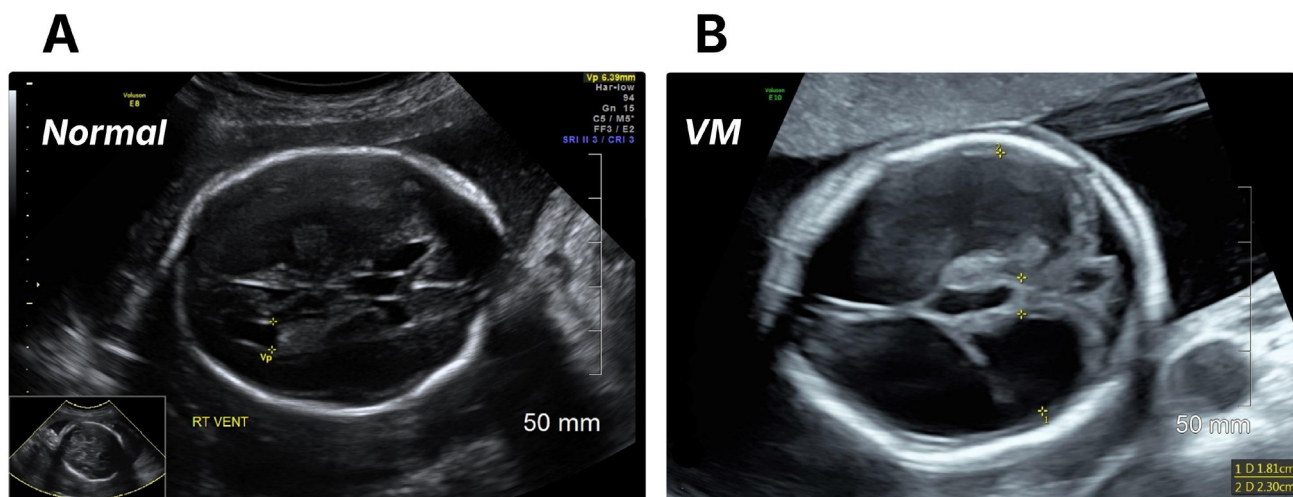


Figure 1. Prenatal ultrasound comparison of normal and ventriculomegaly (VM) fetal brains. (a) Normal fetal brain showing typical lateral ventricle size and configuration [26]. (b) Fetal brain with VM demonstrating enlarged lateral ventricles.

Table 1

Data set composition for ventriculomegaly classification

Subset	Total (n, %)	Normal (n, %)	VM (n, %)
Training	560(68.04%)	462(67.94%)	98(68.53%)
Validation	140(17.01%)	116(17.06%)	24(16.78%)
Test (hold-out)	123(14.95%)	102(15.00%)	21(14.69%)
Total	823(100%)	680(100%)	143(100%)

The data set was divided into training, validation and a held-out test set. The combined training and validation set (85% of the total data) was further partitioned using fivefold cross-validation for model development and hyperparameter tuning, while the test set (15%) was kept completely separate and used only for final performance evaluation.

VM, ventriculomegaly.

the minority class being VM. The VM data set was randomly stratified into training, validation and an independent held-out test set. A total of 560 of the images (68.04%) in each fold were in the training set, 140 of the images (17.01%) were in the validation set and 123 of the images were in an independent hold-out test set, as summarized in Table 1. Stratified sampling ensured that the class-wise distribution was also kept the same: approximately 68% of both normal and VM cases were used for training and 17% for validation, and about 15% of them were kept aside as a final evaluation set. This level of stratification ensured that the majority and minority classes would be represented equally well through all phases of model development.

Data pre-processing

Prior to training the DL models, a multi-stage pre-processing pipeline was applied to all ultrasound images to maintain patient privacy, remove visual artifacts and yield a consistent data set ready for fine-tuning the pre-trained USF-MAE and baseline models. The entire workflow encompassed four major stages: (i) de-identification and scrubbing of the metadata; (ii) removal of on-image annotations and graphical markers; (iii) normalization and resizing of images; and (iv) controlled data augmentation that would assist in generalization for the model. Every step was subjected to proper quality control that would ensure the diagnostically relevant features of the fetal brain were maintained while eliminating confusing visual information.

1) De-identification and scrubbing. Raw ultrasound images frequently contain textual overlays displaying patient information such as name, hospital identification number, examination date or operator initials. These overlays are typically located along the upper border of the scan. To remove such personal health information, we implemented an automated cropping procedure that removed a fixed vertical strip from the top portion of each image. The cropping height was empirically determined after visually inspecting a representative subset of scans to identify the region where textual metadata consistently appeared. All images were then visually checked to ensure that no identifiers remained visible.

This de-identification step ensured both institutional ethics and privacy rules compliance, as well as reduced irrelevant high-contrast text regions that may bias the model during feature learning. The scrubbing script maintained the original folder hierarchy, which facilitated subsequent data management and reproducibility.

2) Annotation and marker removal. In the clinical practice of ultrasound, it is common for operators to annotate images right on the screen with measurement calipers, arrows, text labels or colored highlights that identify anatomical landmarks, as can be seen in Figure 2a. While helpful for human interpretation, these overlays introduce strong, non-biological edges and color signals that may mislead a neural network into associating the markings rather than the anatomy with a specific class label. To mitigate this risk, we applied an automated annotation-removal algorithm similar to the pipeline previously described in our USF-MAE study [25].

Each RGB image was converted to the hue-saturation value color space, where color components are more separable than in standard RGB representation, as shown in Figure 2b. Regions exhibiting unusually high saturation or hue values, characteristic of synthetic markings such as yellow or green text, were detected using adaptive thresholding. The resulting binary mask was refined with morphological operations (dilation followed by closing) to ensure that all annotation pixels and their immediate boundaries were included.

The missing areas were filled in with an inpainting technique based on the Navier-Stokes equations (Fig. 3c) [28]. This method is recognized for its capability to retain texture and intensity details from adjacent zones into the gaps. Following physical principles, it calculates the path of image gradients and prolongs isophote lines smoothly over the masked area, efficiently reconstructing essential echotexture designs. The inpainting made use of OpenCV's version with a three-pixel radius,

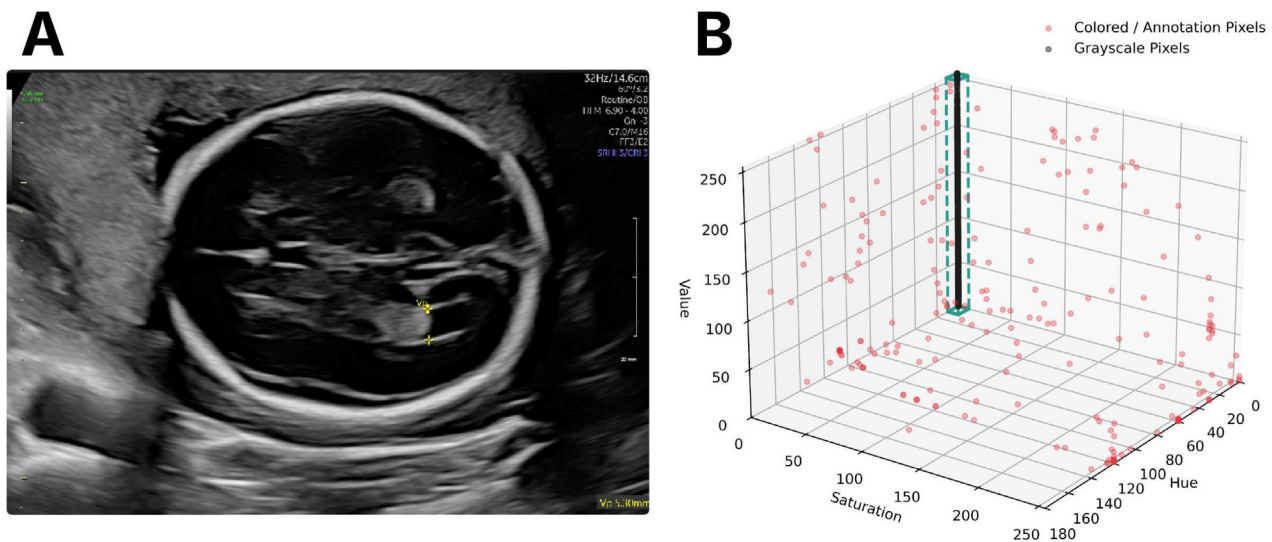


Figure 2. (a) Representative fetal ultrasound image used for analysis [26]. (b) Corresponding pixel distribution mapped in hue-saturation value color space, highlighting valid grayscale pixels versus those outside the grayscale range.



Figure 3. (a) Original fetal ultrasound image. (b) Binary mask identifying non-grayscale artifact regions. (c) Processed image after artifact removal and inpainting using a Navier-Stokes–based [28] inpainting method.

as this provided a compromise between detailed texture completion and speed. Random groups of cleaned images were inspected by eye to ensure that no changes had occurred in the tissue structure, especially in fetal ventricles, and that no invented anomalies had been introduced (see Fig. 3).

3) Image normalization and resizing. After de-annotation, the images were normalized to a uniform spatial resolution of 224×224 pixels. This particular resolution was selected in order to make sure that it would be compatible with the ViT backbone being used in USF-MAE, and also as a trade-off between computational burden and enough structural detail for recognition of ventricular morphology. When resizing, we used nearest neighbor interpolation to avoid generating artificial intermediate pixel values that could blur fine echogenic boundaries.

Pixel intensity values were normalized to match the statistical distribution of the ImageNet data set (mean = [0.485, 0.456, 0.406]; standard deviation = [0.229, 0.224, 0.225]). Ultrasound images are different from general images used in ImageNet, but using the same normalization scheme enables transfer learning of pre-trained models by matching the magnitudes of feature scales in the first layers of the network. This normalization also reduces illumination variance and makes gradient updates more stable during optimization.

4) Data augmentation. To enhance model generalization and control overfitting, a controlled set of geometric data augmentations was applied during training. As the appearance of anatomy in ultrasound is sensitive to the orientation of the probe, scaling and tissue contact pressure, we limited augmentations to transformations that reflect plausible variations, which are encountered within clinical scanning rather than aggressive synthetic distortions. Specifically, each image was subjected to augmentations described as follows:

- Random rotation of the images by an angle in the range 0° – 90° mimicked variation of the probe angle between examinations.
- Flips horizontally and vertically with a probability of 0.5, corresponding to all possible orientations of acquisitions that naturally occur in fetal brain imaging.
- Random resized crop with scale between $0.5 \times$ and $2.0 \times$ the original field of view introduced modest zoom-in and zoom-out variations while keeping the aspect ratio.

Data augmentations were applied at runtime so that the model would see slightly different spatial presentations of the same anatomical pattern for every training epoch. No color or brightness perturbations were applied to preserve grayscale intensity distributions typical in ultrasound images.

Pre-processing and augmentation were carried out with the help of PIL and OpenCV inside a custom PyTorch data-loading pipeline. Therefore, this data set, in its pre-processed version, was finally saved as PNG to keep the image quality intact while reducing file size and ensuring cross-compatibility between operating systems. The standardization procedures ensured a clean, standardized and de-identified data set that

retained all of the clinically relevant information for accurate classification of VM, preventing the model from learning spurious or patient-specific cues.

Model training pipeline

The VM classification model used the USF-MAE [25] encoder fine-tuned on the curated fetal brain ultrasound data set. The encoder was adapted to a binary classification objective for normal versus VM cases (*i.e.*, presence or absence of VM). The training pipeline leveraged the rich representations specific to ultrasound learned during the large-scale self-supervised pre-training phase of USF-MAE. That entire flow included model initialization and fine-tuning with cross-validation, further optimized by adopting an adaptive training schedule. All experiments were run on standardized computational and software configurations to ensure reproducibility.

1) USF-MAE architecture. USF-MAE [25] is a foundation model based on transformers that learns rich and generalizable feature representations from ultrasound data without manual labels. This architecture follows the principles of a masked autoencoder (MAE) first proposed by He et al. in 2022 [29], but has been appropriately tuned for use within the particular domain of ultrasound imaging. The USF-MAE pre-training and fine-tuning of its ViT encoder architectures are presented in Figure 4.

The model employs a ViT backbone as the encoder and a light-weight decoder for image reconstruction. During pre-training, each ultrasound image is divided into fixed-size, non-overlapping patches of 16×16 pixels. A random subset of these patches—25% in our configuration—is masked, and the remaining visible patches are linearly projected into embedding vectors that serve as input tokens for the transformer encoder [25]. The encoder processes only the visible tokens using multi-head self-attention layers to model the spatial relationships among observed regions, thereby learning contextual feature representations that capture both local texture and global anatomical structure.

The decoder then receives a combination of the encoded visible tokens and learnable mask tokens representing the missing patches [25]. With the help of its transformer blocks, the decoder reconstructs the image, including the masked parts, from contextual clues based on the visible areas. The model is pre-trained by minimizing the mean squared error between the reconstructed image $\hat{x}$ and the original image x , encouraging the encoder to learn modality-specific priors inherent to ultrasound signals, such as speckle texture, acoustic shadowing and tissue boundary patterns. The reconstruction loss is defined as eqn (1):

$$\mathcal{L}_{MAE} = \frac{1}{N} \sum_{i=1}^N \|\hat{x}_i - x_i\|_2^2, \quad (1)$$

where N is the number of pixels in the image, x_i denotes the original pixel intensity and $\hat{x}_i$ denotes the reconstructed pixel intensity.

USF-MAE was pre-trained on $\sim 370,000$ ultrasound images aggregated from 46 open-source data sets (OpenUS-46) covering 23

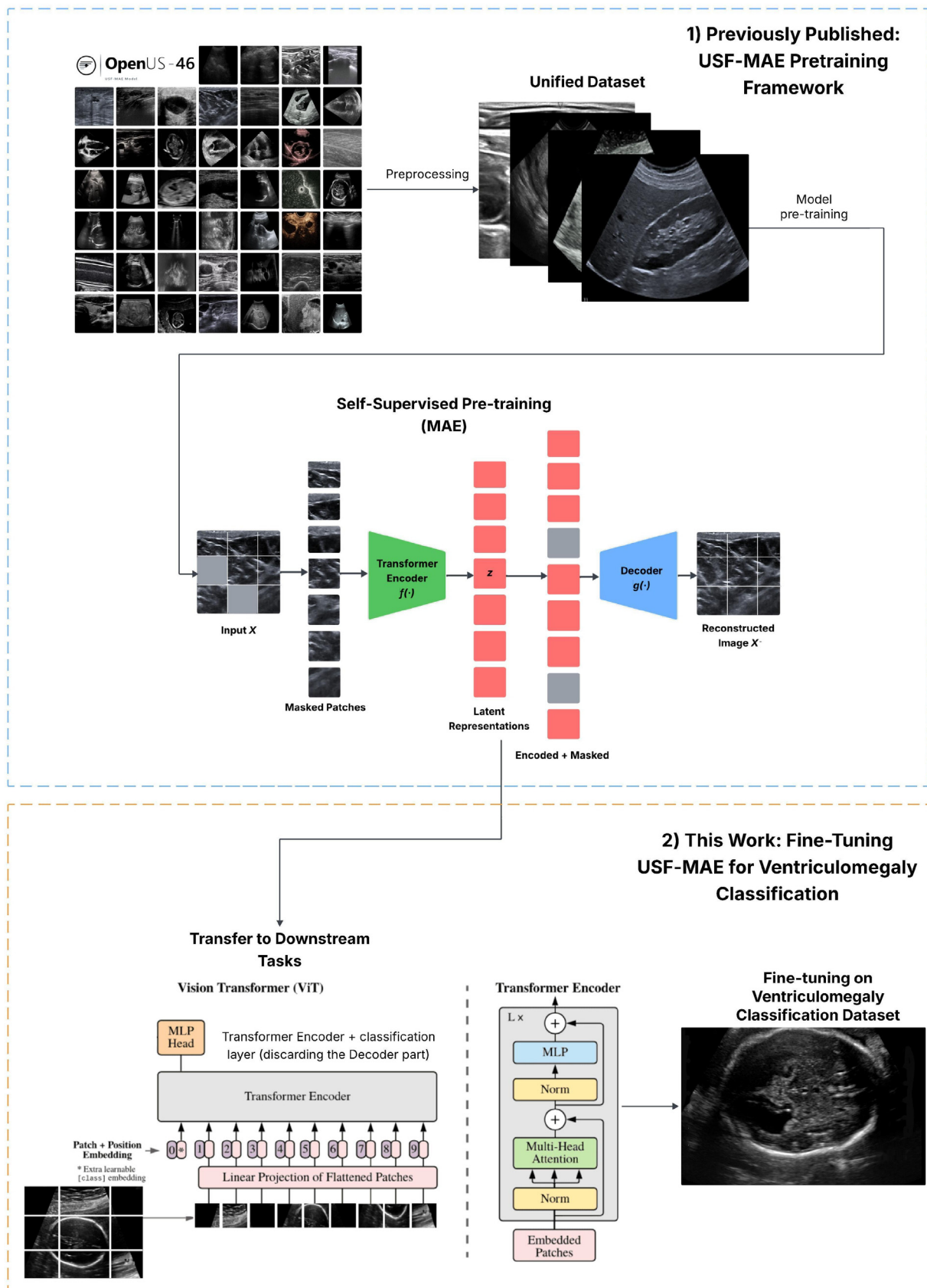


Figure 4. Overview of the proposed ventriculomegaly (VM) classification pipeline. (a) Previously published ultrasound self-supervised foundation model with masked autoencoding (USF-MAE) pre-training framework: A unified ultrasound data set undergoes pre-processing and is used for self-supervised pre-training with a masked autoencoder (MAE), where random patches are masked (25% masking ratio) and reconstructed to learn ultrasound-specific latent representations [25]. (b) This work: The pre-trained encoder is transferred to the downstream VM classification task by discarding the MAE decoder and attaching a classification head. The model is then fine-tuned on the labeled VM data set to enable clinical prediction.

anatomical regions (Fig. 4) [25]. Such large-scale, self-supervised pre-training helps to build up a holistic representation of ultrasound features over several organ systems within the encoder; hence, a promising foundation to then fine-tune to new downstream tasks requiring specialization for diagnosis, such as the detection of VM in our study.

2) Model initialization. The starting point for the experiments was the USF-MAE pre-trained encoder, which had been previously trained on ~370,000 ultrasound images from the OpenUS-46 corpus. The pre-trained encoder provided a robust initialization for ultrasound image representation, allowing the model to generalize efficiently from unlabeled domain data to this specialized diagnostic task.

For this study, the MAE decoder module, responsible for reconstructing masked patches during pre-training, was removed and replaced with a task-specific classification head. The new head consisted of a fully connected (dense) layer mapping the encoder's latent feature vector to two output nodes, corresponding to the binary classes: normal ventricles and VM. A SoftMax activation function was applied to convert logits into class probabilities. The rest of the encoder weights were retained and fine-tuned jointly with the classification head to enable end-to-end optimization.

3) Fine-tuning process. To ensure unbiased evaluation and avoid data leakage, the data set was divided into an independent hold-out test set and a training/validation set, as described in the 'Data set distribution and splitting strategy' section. The independent test set, consisting of 15% of images from each class, was held out before model development and used only for final performance assessment. After completing the fivefold cross-validation on the remaining 85% of the data, each fold-specific checkpoint was applied to the independent held-out test set, and the resulting metrics were averaged across folds to obtain a robust estimate of test performance. This approach was adopted to enhance model generalization and mitigate the effects of class imbalance. The data set has a 1:5 ratio between VM and normal cases (Table 1). To ensure fair learning across both classes, the data (training and validation subsets) were randomly stratified into five folds, maintaining the same proportion of classes in each subset. Training was done on four out of five splits in every iteration (i.e., 80% of the data set), with the remaining split being used as an internal validation set. The results obtained after running all of the splits were averaged out to get mean values for validation metrics.

Given the class imbalance, we applied class weighting within the cross-entropy loss function during model fine-tuning. Weights of classes were calculated inversely proportional to class frequencies in the training folds, thus effectively penalizing the model more for any misclassification on the positive class (VM). This leads to a weighting strategy that drives the model's focus more on minority classes during optimization steps and hence tunes sensitivity toward positive detections without losing total accuracy.

The pre-processing and augmentation pipeline, as described in the 'Data pre-processing' section of the Methodology, was applied here in each fold. Data were loaded batchwise and pre-processed on-the-fly with the help of PyTorch's data loaders, thus ensuring consistent augmentations as well as memory efficiency.

4) Optimization and hyperparameter configuration. The model was fine-tuned and optimized using AdamW [30], an optimizer that incorporates Adam's adaptive moment estimation with decoupled weight decay regularization. It is known to maintain stability and convergence efficiency during fine-tuning of transformer-based architectures on medical image data.

An exhaustive grid search over batch size and learning rate within the below-stated ranges was conducted to try to identify the best training configuration: batch size $\in \{32, 64, 128\}$, learning rate $\in \{0.001, 0.0005, 0.0003, 0.0001\}$ and weight decay $\in \{0.01, 0.05, 0.001, 0.0001\}$. A fivefold cross-validation was run for every combination, and the one that produced the highest mean accuracy across folds was used. The optimal setup was found to be: learning rate = 0.0003, weight decay = 0.0001 and batch size = 128.

Cosine annealing with linear warm-up scheduled the learning rate such that it increased linearly in the first 10% of total training epochs (warm-up), then decayed smoothly following a half-cycle cosine curve. Such scheduling does not allow the model to converge prematurely and enables stable training at relatively high learning rates. Gradient clipping by maximum norm 1.0 was also applied as a stabilizing weight update and to avoid exploding gradients, common to neural networks.

Every training epoch comprised forward and backward passes over the data set and was thereafter followed by an evaluation step on the validation subset. The model's best state within each fold, validated by the highest possible validation accuracy, was saved as a checkpoint to prevent degradation of performance due to overfitting. The resulting performance metrics were averaged across the best checkpoints from all the folds.

5) Hardware and software configuration. Experiments ran in PyTorch, and hugging face transformers were used to load, train and infer models. Data manipulation and evaluation used NumPy, Pandas and scikit-learn.

Training was performed on a high-performance SLURM-managed GPU cluster at the hospital research institute. Each run utilized one NVIDIA L40S GPU with 128 GB of the overall system's RAM. Each fold for every combination of hyperparameters took about 45 min at 100 epochs. Model checkpoints and logs were saved so that results could be tracked and reproduced across different experiments.

The USF-MAE foundation model weights and the OpenUS-46 pre-training corpus are publicly available and can be downloaded from our project repository (GitHub repository: <https://github.com/Yusufii9/USF-MAE>).

Baseline models for comparison

A benchmark was set through the selection of three well-known DL architectures as baselines. This included VGG-19 [31], ResNet-50 [32] and vision transformer base (ViT-Base or ViT-B/16) [33]. These models are considered standard in the evolution of DL architectures, from convolutional feature extractors to transformer models capable of global attention modeling. They were also trained under the same pre-processing, data augmentation and cross-validation protocols as used for the USF-MAE fine-tuning framework, so that a direct comparison could be made.

- 1) VGG-19: VGG-19 [31] is a type of DL model known for its depth and complexity. The VGG-19 model is composed of 3×3 convolutional kernels padded with maximum pooling layers and followed by two fully connected layers and one final classification layer representing some purely deep architectures with a fairly simple structure consisting of uniformly small-kernel convolutions that allow different levels of spatial features abstraction. For this study, it was initialized with pre-trained weights from ImageNet-1K (~1.2 million training images over 1000 classes) fine-tuned end-to-end on the VM data set using a batch size of 64, learning rate of 0.0003, weight decay of 0.05, the best hyperparameters found and trained for 100 epochs in order to return logits corresponding to the binary classes (normal vs. VM) after adapting the classification head of the model architecture appropriately. VGG-19 does not include any residual connections or attention mechanisms; however, the manner in which it has such a deep convolutional stack offers strong baselines for evaluating classic feature hierarchies in ultrasound imagery.
- 2) ResNet-50: The ResNet-50 [32] model implements residual learning using skip connections to effectively solve the problem of vanishing gradients in very deep networks. Each residual block learns a mapping that is an addition to its input, making it easier for the network to learn incremental changes rather than relearning everything from scratch. The architecture results in high representational capacity with efficient gradient propagation through

layers. In this work, we fine-tuned ResNet-50 that was initialized with ImageNet-1K pre-trained weights using a batch size of 64, a learning rate of 0.001 and a weight decay of 0.05 for 100 epochs. The final fully connected layer was updated such that it was replaced by a binary classifier specific to the VM detection task. The ResNet-50 provides a convolutional baseline and good validation performance on numerous medical imaging applications.

- 3) ViT-Base (or ViT-B/16): The ViT-B/16 [33] model represents the transformer paradigm for visual representation learning. Rather than using convolutions, it breaks every input image into 16×16 non-overlapping patches linearly embedded as tokens added by positional encoding. These are fed through several layers of multi-head self-attention and feed-forward blocks: as dependencies at long distances and global context can be picked up by the model, it should be able to understand ultrasound images where structural cues are spread spatially across an image. Notably, ViT-B/16 shares the same backbone architecture as the encoder used in USF-MAE; the key difference lies in the pre-training strategy (ImageNet-1K supervised pre-training for ViT-B/16 vs. large-scale self-supervised ultrasound domain pre-training for USF-MAE). The ViT-B/16 model is initialized from ImageNet-1K pre-trained weights and uses a batch size of 64 and a learning rate equal to 0.0003, with weight decay set to 0.01 for 100 epochs. Its classification head was changed to provide two logits for the binary problem instead of the ImageNet-1K 1000 classes.
- 4) MoCo v3 (self-supervised ViT): To disentangle the benefits of self-supervision from those of ultrasound-specific pre-training, we included an additional baseline using the MoCo v3 framework [34] adapted to a ViT (ViT-B/16) architecture. MoCo v3 is a self-supervised learning (SSL) approach where representations are learned by matching differently augmented views of the same image using momentum-updated encoders. We initialized the ViT-B/16 backbone with their publicly available SSL weights (ViT-B pre-trained for 300 epochs), before fine-tuning for binary VM classification under the same training protocol as the other models. This baseline enables comparison between generic self-supervised pre-training and the ultrasound-specific pre-training of USF-MAE.
- 5) Evaluation conditions: Baseline models were fine-tuned on the same fivefold partition for cross-validation and held-out test set, and with the same pre-processing plus data augmentation steps described previously. The training epochs, batch sizes and optimization schedules were kept alike in all experiments so that any differences seen would reflect the strength of each design rather than changes in the training method.

Together, these starting points offer a wide mix of design ideas, from known convolutional models to new transformer architectures, against the performance and generalization ability of the proposed USF-MAE architecture, allowing it to be critically evaluated.

Performance metrics

The performance of the proposed USF-MAE model and all baseline architectures was quantified using several metrics that collectively characterized classification accuracy, diagnostic sensitivity and discriminative power. As this work is ultimately designed for a clinical application where verified detection of VM has important implications, metrics that quantitatively describe the ability to correctly identify positive (VM) cases while simultaneously minimizing false negatives have been emphasized. All metrics were computed on both the validation and independent held-out test sets, with the final results being their mean values over the five cross-validation folds.

1) Accuracy. Accuracy was defined as the total proportion of correctly classified samples out of all predictions made, defined as eqn (2):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

Where TP is true positives, TN is true negatives, FP is false positives and FN is false negatives. Accuracy gives a general indicator of performance, but accuracy alone cannot fully describe the quality of the model when applied to imbalanced data sets.

2) Precision. Precision (positive-predictive value) is the ratio of true-positive predictions to all positive predictions made by the model. A high-precision value for this model would mean that it does not frequently label normal cases as having VM; hence, it is accurate when it makes a positive prediction, which is an important class imbalance metric. It is expressed as eqn (3):

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

Clinically, precision has great relevance in this situation, because false-positive VM diagnoses lead to unwarranted parental anxiety and other unnecessary follow-up tests.

3) Recall (sensitivity). Recall or sensitivity is defined as the proportion of actual positive cases that are correctly identified by the model, given by eqn (4):

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

High recall means a strong ability to identify the real cases of VM with a low chance of missing any diagnosis. This is a major attribute in any clinical screening scenario.

4) F_1 -score. The F_1 -score is the harmonic mean of precision and recall, balancing the trade-off between the two. It is particularly informative for imbalanced data sets, where accuracy can be misleading. The F_1 -score is defined as eqn (5):

$$F1 - \text{score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (5)$$

A high F_1 -score speaks to strong general performance, capturing both the model's capacity to pick out real cases and its ability to avoid false alarms.

5) Specificity. Specificity, or the true negative rate, details the model's strength at correctly tagging normal (non-VM) cases. It is calculated as eqn (6):

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (6)$$

Keeping specificity high is key when used in clinics as it lowers the chance of wrongly saying healthy fetuses have VM.

6) The area under the receiver operating characteristic curve. The area under the receiver operating characteristic curve (AUC-ROC) also gives a threshold-independent measure of how well the model can differentiate between positive and negative classes. Recall at all thresholds plotted against $1 - \text{specificity}$ gives the ROC curve; that is, recall (the true-positive rate) for all possible decision thresholds versus false-positive rates, or $1 - \text{specificity}$. AUC values lie in the range $[0, 1]$, where 0.5 is equivalent to random guessing and 1 is perfect discrimination. More specifically, higher AUC values mean that for most comparisons between a true VM case and a normal case, the model assigns higher scores to true VM cases.

Results

Tables 2 and 3 report a summary of the comparative performance of all models evaluated on the cross-validation and the independent held-out test sets, respectively. The models considered in this comparison include the two convolutional baselines (VGG-19 and ResNet-50), the

Table 2

Comparative performance of baseline and ultrasound self-supervised foundation model with masked autoencoding models on ventriculomegaly classification (validation set)

Model	Validation set (%)					
	Accuracy	Precision	Recall	F_1 -score	Specificity	AUC
VGG-19	92.14	86.27	66.10	72.39	97.58	88.54
ResNet-50	96.43	92.83	86.80	89.45	98.44	95.23
ViT-B/16	95.57	90.94	83.57	86.73	98.09	95.72
MoCo v3	95.43	89.25	84.43	86.59	97.75	94.86
USF-MAE (ours)	97.29	96.55	87.67	91.76	99.31	96.00

AUC, area under the curve; USF-MAE, ultrasound self-supervised foundation model with masked autoencoding.

Bold values denote the highest value achieved for each evaluation metric, highlighting the best-performing model for that metric among all models compared in the table.

Table 3

Comparative performance of baseline and ultrasound self-supervised foundation model with masked autoencoding models on ventriculomegaly classification (test set)

Model	Test set (%)					
	Accuracy	Precision	Recall	F_1 -score	Specificity	AUC
VGG-19	92.68	84.70	69.52	75.63	97.45	88.43
ResNet-50	96.26	88.25	90.48	89.22	97.45	97.50
ViT-B/16	93.82	88.59	74.29	79.85	97.84	97.00
MoCo v3	94.80	84.98	84.76	84.79	96.86	97.41
USF-MAE (ours)	97.24	94.47	89.52	91.78	98.82	97.00

AUC, area under the curve; USF-MAE, ultrasound self-supervised foundation model with masked autoencoding.

Bold values denote the highest value achieved for each evaluation metric, highlighting the best-performing model for that metric among all models compared in the table.

transformer baselines (ViT-B/16 and MoCo v3), as well as the proposed USF-MAE model, which is fine-tuned for VM classification.

On the validation set (Table 2), USF-MAE demonstrates the highest performance with respect to all the evaluation metrics, reaching an accuracy of 97.29%, a precision of 96.55%, a recall of 87.67%, an F_1 -score of 91.76%, a specificity of 99.31% and an AUC of 96.00%. These results outperformed the corresponding values of ViT-B/16 (accuracy: 95.57%, F_1 -score: 86.73%), while the self-supervised MoCo v3 model reached an F_1 -score of 86.59%, with ResNet-50 achieving an accuracy of 96.43% and an F_1 -score of 89.45%. Although MoCo v3 outperformed the supervised ViT-B/16 baseline in recall (84.43% vs. 83.57%), it did not surpass USF-MAE, indicating that ultrasound-specific pre-training provides additional benefits beyond generic SSL alone. While the observed validation AUC of 96.00% points toward strong overall separability between classes, it needs to be interpreted with care in the context of the class imbalance in the data set. The F_1 -score (91.76%), the recall (87.67%) and the precision (96.55%) metrics, on the other hand, confirm that USF-MAE retains high detection rates of the minority (VM) class and is able to produce few false negatives on this data set.

The generalization of the model to an independent test set was evaluated by applying each of the five cross-validation checkpoints to the held-out test data and averaging the resulting performance metrics across folds (Table 3). This cross-fold evaluation averaging provides a robust estimate of the model's generalization capability. As can be seen, the proposed USF-MAE model retained its superior generalization under this evaluation protocol, yielding an accuracy of 97.24%, a precision of 94.47%, an F_1 -score of 91.78% and a specificity of 98.82%, while maintaining high recall (89.52%). Although the recall of ResNet-50 (90.48%) was marginally higher, its overall precision and F_1 -score were inferior to those of USF-MAE, pointing toward a possible trade-off between true-positive rate and confidence in predictions. The AUC of 97.00% achieved

by USF-MAE on the test set points toward the model's ability to robustly distinguish VM from normal cases under unseen clinical conditions.

Collectively, these results show that USF-MAE outperformed CNN and generic ViT baselines, including a self-supervised transformer model (MoCo v3), that were pre-trained with ImageNet-1K weights for VM detection from fetal ultrasound images. The observed improvement is most likely due to its pre-training in a large-scale SSL task on the OpenUS46 corpus [25], which results in the learning of ultrasound-specific textural and anatomical features that transfer well to the downstream task of prenatal imaging. The high precision and specificity of USF-MAE also point toward a reduction in false-positive diagnoses, which is a highly desirable property in a clinical setting where minimizing parental anxiety and reducing the burden of follow-up testing are important.

In summary, the results on the validation and test sets confirm that USF-MAE is a reliable and label-efficient foundation model for ultrasound-based fetal brain anomaly classification, achieving state-of-the-art performance even with limited labeled data.

Discussion

Interpretation of results and model performance

Across both the validation and the held-out independent test sets, the USF-MAE model outperformed all baseline models, as indicated in Tables 2 and 3. As shown in Figure 5 (a–d), the ROC curves and precision-recall curves from the fivefold ensemble had very narrow standard deviations, with the mean AUC values consistently greater than 0.96 for all fivefolds (held-out independent test set). This suggests that there was little variance in performance across the folds, and that the proposed model was able to generalize well despite the relatively small amount of data, as well as the class imbalance between normal and VM cases.

On the held-out test set, the ROC curves in Figure 5c demonstrated that almost all of the true VM cases were correctly classified, with a mean AUC of 0.97 ± 0.005 . The mean precision-recall curve in Figure 5d had an area of 0.95 ± 0.011 , which showed that the model was able to maintain both high recall and high precision across all decision thresholds. These findings are aligned with clinical priorities, as recall was of particular importance to ensure that there were few missed true-positive VM cases.

In addition, the confusion matrix in Figure 6 shows that there were only 17 mis-classifications across the entire aggregated mean test cohorts (6 false positives and 11 false negatives). The model had a specificity of 98.8% by correctly classifying 504 out of 510 normal cases, and a recall of 89.5% by identifying 94 out of 105 VM cases. We considered this trade-off between recall and specificity to be appropriate for a prenatal screening setting, where over-diagnosis could lead to unnecessary parental anxiety but under-diagnosis could result in delayed further investigation of potential brain abnormalities. The robustness and consistency of these performance metrics across the validation and independent held-out test sets provided strong evidence to support the reliability of the USF-MAE encoder as a foundation model that could be applied to generalize to other unseen limited fetal ultrasound data sets.

These results overall demonstrated that the domain-specific pre-training on the large-scale OpenUS46 corpus gave the proposed USF-MAE model a significant advantage over conventional CNNs and generic ViTs. By pre-training the network to learn ultrasound-specific representations that captured both the global anatomical context and the fine-grained echotextural features, the model was able to achieve superior diagnostic performance using a minimal amount of labeled data. This highlights the potential of self-supervised foundation models to enable data-efficient clinical implementation in obstetric ultrasonography.

Explainable artificial intelligence visualization

Explainability is a critical component in the clinical translation of DL models, particularly in high-stakes domains such as prenatal imaging.

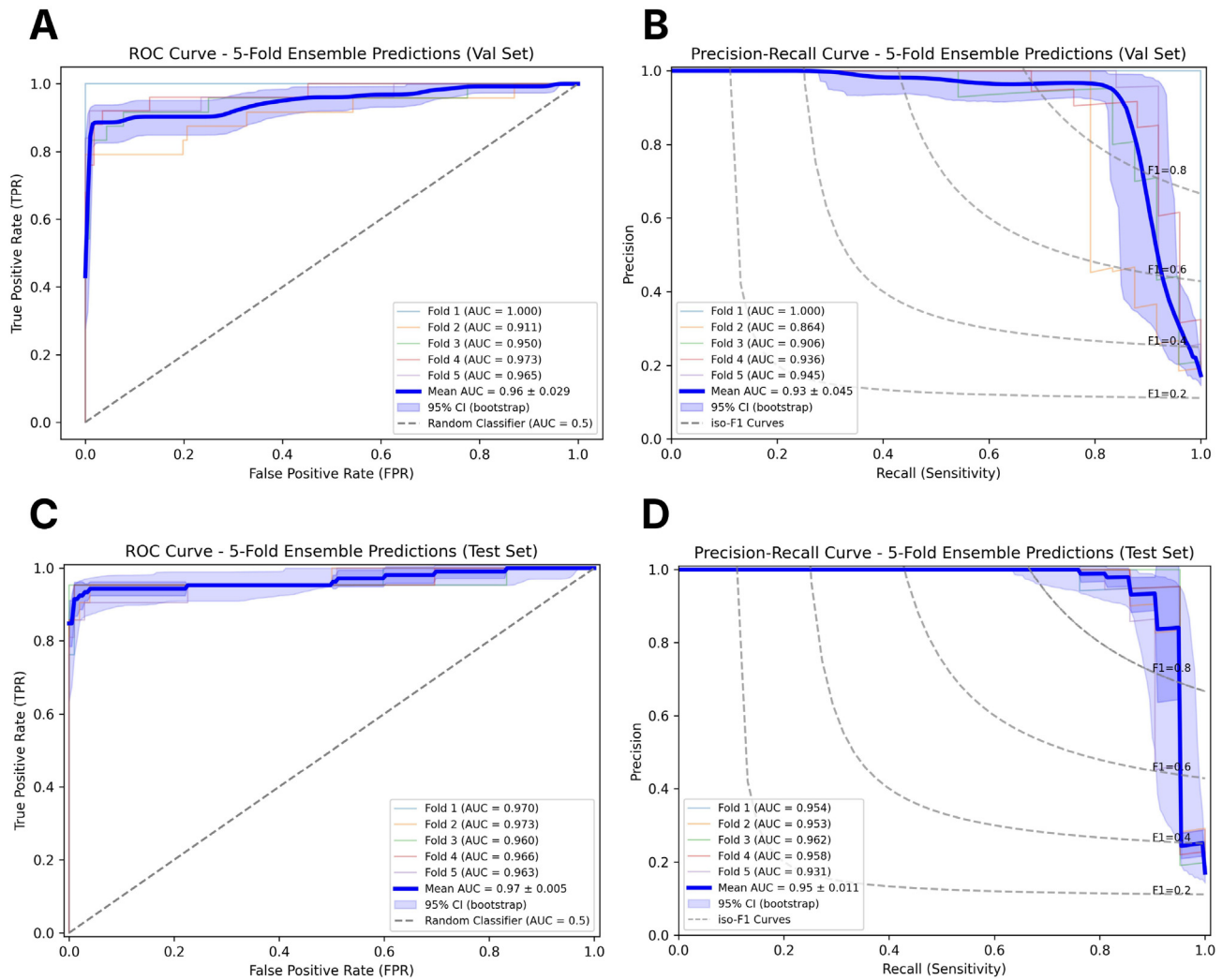


Figure 5. (a, c) Receiver operating characteristic and (b, d) precision-recall curves for the fivefold ensemble model on the validation (a, b) and test (c, d) sets. Curves for individual folds are shown along with the mean ensemble performance and variability across folds. AUC, area under the curve; CI, confidence interval.

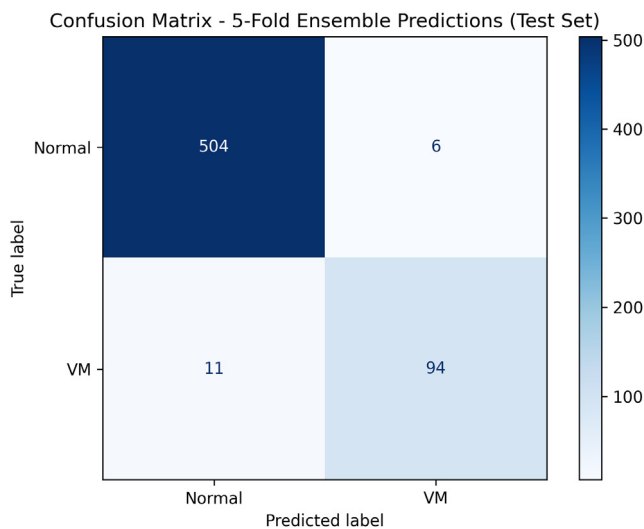


Figure 6. Confusion matrix for the fivefold ensemble model on the test set, showing true versus predicted class labels for normal and ventriculomegaly (VM) samples.

To interpret the decision-making process of the fine-tuned USF-MAE model, we employed both Eigen-CAM [35] and Grad-CAM [36] methods, as visualized in Figure 7. Eigen-CAM provides class activation maps (CAMs) based on the principal components of the model's activation space, while Grad-CAM leverages gradient information to highlight discriminative regions contributing to the model's predictions.

In the top row of Figure 7 (a–c), the model correctly focused on the periventricular region and the boundaries of the lateral ventricles for normal cases. In contrast, the bottom row of Figure 7 (d–g) demonstrates that, for VM cases, the activation maps concentrated along the dilated ventricular cavities, with high-intensity regions corresponding to the expanded atrial width, the primary sonographic hallmark of VM. The smooth and spatially consistent activation distribution observed across all examples suggests that the model learned to recognize structural features relevant to fetal brain morphology rather than relying on spurious artifacts or background patterns.

While several gradient-based visualization techniques exist, including Grad-CAM [36] and its variants, Eigen-CAM [35] offers distinct advantages when applied to ViT-based architectures such as USF-MAE. Grad-CAM depends on spatial gradients propagated through convolutional feature maps to localize discriminative regions, a mechanism well-suited to CNNs but less effective for transformers, which operate on patch tokens without explicit convolutional kernels. In contrast, Eigen-CAM derives its visualizations directly from the dominant eigenvectors of the attention-based feature

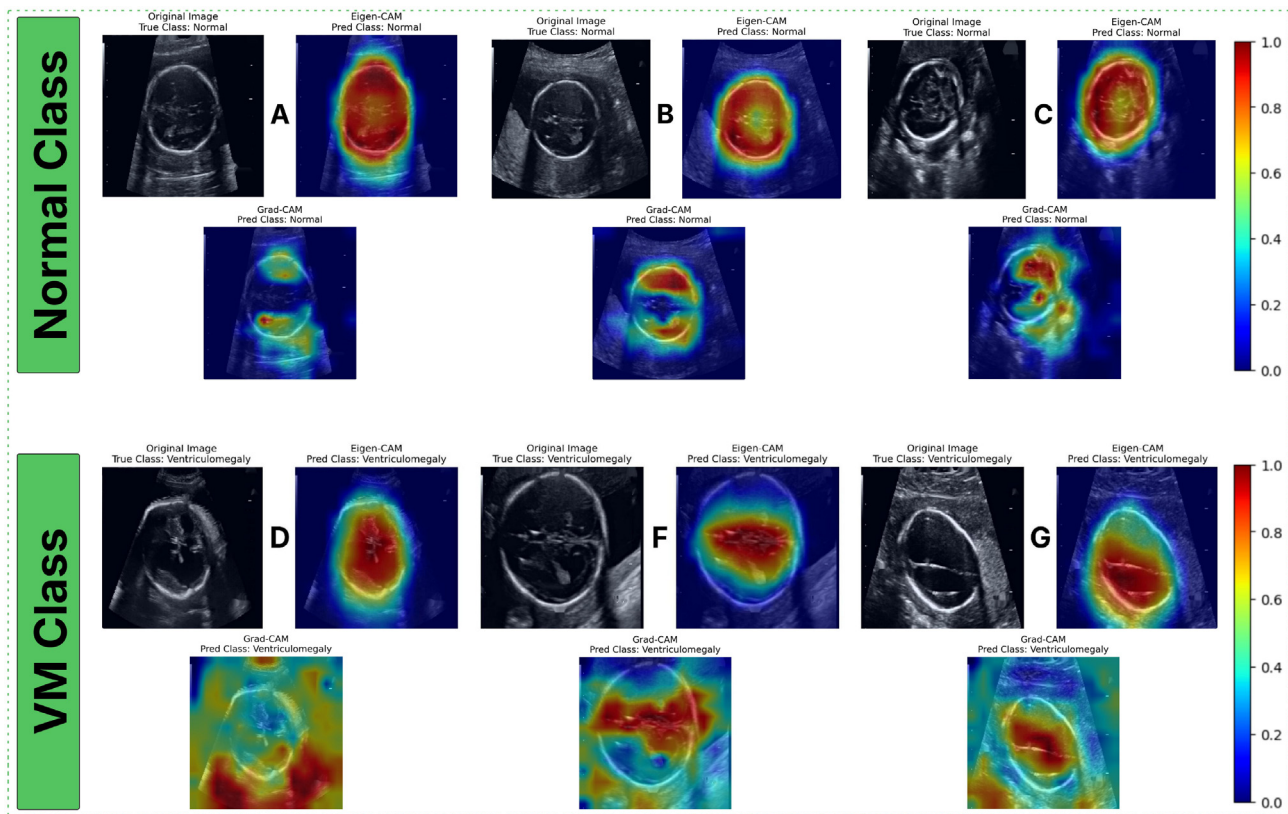


Figure 7. Representative ultrasound images and corresponding Eigen-CAM and Grad-CAM heatmap visualizations for correctly classified cases from both classes. Columns show original images and the model's highlighted regions used for prediction. (a–c) Normal class examples. (d–g) Ventriculomegaly (VM) class examples.

activations, capturing the principal modes of variance across all attention heads without gradient computation. This gradient-free formulation preserves the global context inherent to transformer attention mechanisms and avoids the instability or noise amplification that can arise during gradient backpropagation.

To further validate the robustness of the interpretability analysis, Grad-CAM visualizations were included alongside Eigen-CAM in Figure 7. While Grad-CAM was able to highlight broadly relevant regions, its activations were often more spatially diffuse and less anatomically constrained compared with Eigen-CAM. In several examples, Grad-CAM emphasized multiple scattered regions within the cranial cavity, whereas Eigen-CAM consistently produced smoother and more localized attention around the ventricular system. This difference is consistent with the architectural characteristics of transformer-based models, where gradient-based localization may be less stable due to the lack of explicit spatial hierarchies. The complementary use of both methods demonstrates that, despite methodological differences, the model consistently attends to clinically meaningful regions, thereby strengthening confidence in the learned representations.

Consequently, Eigen-CAM produced smoother, anatomically coherent saliency maps that aligned closely with the true regions of clinical interest—in this case, the ventricular system. As shown in Figure 7, the USF-MAE attention heatmaps unambiguously identified the ventricular edges in normal and VM cases. Both Eigen-CAM and Grad-CAM visualizations indicate that the model makes its predictions based on physiology-related features. This adds to model transparency and clinical trustworthiness, both of which will be important for eventual regulatory approval and real-world integration into obstetric workflows.

Analysis of model failure cases

To better understand the limitations of the proposed framework and to improve transparency, representative failure cases were analyzed

using both Eigen-CAM and Grad-CAM visualizations. Figure 8 presents two FP and two FN examples from the independent test set, together with the corresponding attention maps generated by the USF-MAE model.

In the false-positive cases (Fig. 8a, 8b), the model incorrectly classified normal fetal brain images as VM. Inspection of the Eigen-CAM heatmaps revealed that the model focused on the overall intracranial cavity and periventricular regions rather than sharply localized ventricular boundaries. In these examples, the ventricular atrium appeared subjectively prominent due to probe orientation or acoustic shadowing, which may have led the model to over-estimate ventricular dilation. Such cases reflect scenarios in clinical practice where borderline ventricular appearance or sub-optimal imaging planes can mimic VM. In addition, Grad-CAM visualizations for these cases often exhibited broader and less localized activations, sometimes extending beyond the ventricular boundaries into surrounding brain tissue. This suggests that, under sub-optimal image quality or ambiguous anatomical presentation, the model may rely on coarse global intensity or shape cues rather than precise ventricular delineation. Such behavior highlights a sensitivity to image quality factors such as acoustic shadowing, speckle noise and off-axis acquisition, which can distort the perceived ventricular morphology.

In the FN cases (Fig. 8c, 8d), the model failed to identify VM and instead predicted a normal class. We observed that Eigen-CAM heatmaps from these failure examples appeared to attend more globally across the brain surface rather than locally around the ventricular atrium. Qualitatively, the ventricles also tended to be smaller/more asymmetrical in these examples, or appeared to have less visually obvious VM. These examples suggest that the model may fail to capture more subtle/marginal VM. Grad-CAM maps in these FN cases further revealed inconsistent or fragmented attention patterns, indicating reduced confidence in identifying the ventricular region. These findings suggest that mild or borderline VM cases, where ventricular enlargement is subtle or asymmetric, may not produce sufficiently distinctive features for the

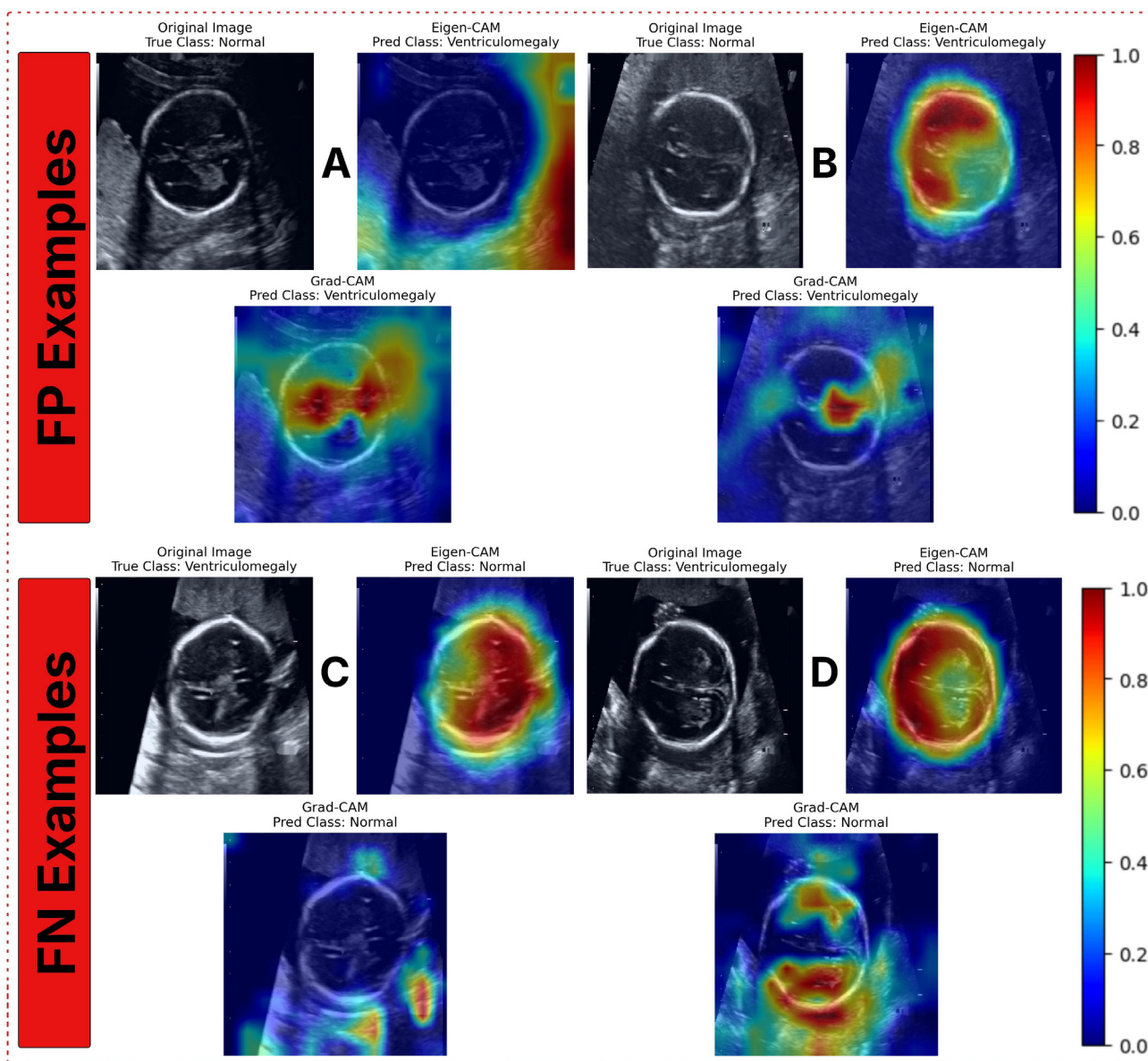


Figure 8. Representative failure cases of the proposed ultrasound self-supervised foundation model with the masked autoencoding model with corresponding Eigen-CAM and Grad-CAM visualizations. Top row: False-positive (FP) examples, where normal fetal brain images were incorrectly classified as ventriculomegaly (VM). Bottom row: False-negative (FN) examples, where VM cases were incorrectly classified as normal. Columns show the original ultrasound image and the associated Eigen-CAM and Grad-CAM heatmaps, highlighting regions that contributed to the model's prediction.

model to reliably differentiate from normal anatomy. Additionally, variations in fetal position, partial volume effects and reduced image contrast may further obscure ventricular boundaries, contributing to missed detections.

In general, failure cases tended to involve challenging imaging conditions, including borderline ventricular enlargement, sub-optimal acquisition plan low contrast and noise artifacts. These factors can obscure ventricular boundaries or exaggerate their appearance, leading to ambiguity in both human and model interpretation. However, the Eigen-CAM maps in these incorrect examples still validate that the model mostly attends to brain structures and not background noise.

We can use these failure cases to guide future work. For example, we may want to consider using multiple views, cine-loops or supervised regression of ventricle size to make the model more sensitive to mild VM while preserving its high specificity. We may also use these cases to build uncertainty-aware or human-in-the-loop systems that refuse to make predictions on such ambiguous examples.

Model complexity and inference efficiency

The computational footprint of each model is summarized in Table 4. Although VGG-19 had the largest parameter count (139.6M), its

Table 4

Trainable parameters and inference time for each model on the independent test set by taking the mean of fivefold models

Model	Parameters (M)↓	Inference time (s) ↓
VGG-19	139.6	10.7
ResNet-50	23.5	8.3
ViT-B/16	85.8	4.3
MoCo v3	85.8	11.3
USF-MAE (ours)	85.8	5.7

Inference time is measured as total wall-clock time to run inference over the entire test set (123 images) using a single NVIDIA RTX 4080 Super GPU.

inference time on the held-out test set (123 images) was 10.7 s on an NVIDIA RTX 4080 Super GPU. ResNet-50 (s), despite having substantially fewer parameters (23.5 M), required 8.3 s. Among transformer-based models, ViT-B/16 achieved the fastest inference time (4.3 s), while USF-MAE required 5.7 s and MoCo v3 required 11.3 s. As ViT-B/16, MoCo v3 and USF-MAE share the same ViT-B/16 backbone size (85.6 M parameters), these latency differences primarily reflect implementation and runtime effects rather than architectural scale. Overall, the measured inference time for USF-MAE corresponded to approximately 46 ms per image on the RTX 4080 Super GPU, supporting the feasibility of near real-time deployment in clinical workflows under GPU-enabled settings.

Generalizability across ultrasound manufacturers and point-of-care devices

While we showed that the proposed USF-MAE model learned visual features that generalize well to fetal brain ultrasound images acquired from GE Voluson systems, model performance could suffer if applied to images from ultrasound scanners from other vendors (Philips, Samsung, Siemens, etc.) or acquired from hand-held/point-of-care ultrasound devices that are becoming more prevalent in low-resource and remote clinical environments. Model performance would likely degrade somewhat due to domain shift caused by differences in transducer technology and beamforming, differences in image post-processing pipelines or image quality, varying dynamic range and speckle statistics between vendors, or differences in image compression artifacts between vendor/software and device class (cart vs. hand-held).

Hand-held/point-of-care ultrasound devices typically have lower spatial resolution, signal-to-noise ratio and penetration depth compared with clinical cart-based ultrasound systems. We acknowledge that this might conceal the subtle borders of fetal ventricles and decrease the fine echotextural features that our model may leverage, affecting model sensitivity or causing false positives when applied to images acquired differently from the acquisition types in the training set.

Despite these challenges, the ultrasound-specific, self-supervised, pre-training strategy employed by USF-MAE is expected to offer improved robustness relative to conventional CNNs and ViTs pre-trained on natural images. By pre-training on a large and diverse corpus of ultrasound data spanning multiple anatomical regions and acquisition conditions, USF-MAE learns modality-specific representations that emphasize fundamental ultrasound characteristics, such as speckle patterns, tissue interfaces and structural geometry, rather than device-specific image appearance. This property likely contributed to the improved generalization observed in this study compared with pre-trained ImageNet baselines.

However, one limitation of our study was that cross-vendor or cross-device generalization was not directly tested in the current work. We plan to externally validate our framework on multi-vendor data sets and images obtained from hand-held ultrasound devices in future studies. We are also interested in domain adaptation or light-weight fine-tuning with a limited amount of labeled data from new devices. That said, the USF-MAE model itself was previously evaluated across three distinct clinical applications [25], each involving different ultrasound machines and acquisition centers, providing preliminary evidence of its robustness across hardware and institutional variations.

Additionally, because the training and test data were derived from a retrospective clinical repository rather than a prospective trial, no formal multi-reader, inter-observer variability analysis was performed at the time of image acquisition or annotation. This limits our ability to quantify label consistency across sonographers or interpreting physicians.

Finally, while correlating artificial intelligence predictions with longitudinal outcomes, such as postnatal neuroimaging or decisions surrounding pregnancy management, would enhance clinical validity, this study was designed as a technical evaluation of artificial intelligence performance relative to expert-labeled prenatal ultrasound data. As such, outcomes-linked adjudication of false-positive and false-negative

cases falls outside the defined scope but remains an important direction for future work.

Furthermore, the absence of demographic details such as maternal age, ethnicity and gestational age distribution, along with an assessment of how these align with regional birth cohorts, limits the ability to evaluate the external validity of the model across diverse patient populations.

Future work and clinical implications

Despite the robust diagnostic performance observed with the current model, there are several avenues for future work that can be explored to prepare for clinical integration. Expanding the training corpus with data from other institutions and ultrasound systems could be explored to evaluate the model's robustness across different devices, operators and patient populations. Furthermore, incorporating 3-D ultrasound volumes or cine-loop sequences of the fetal brain could be considered to allow for the learning of temporal features, which may be more sensitive to subtle changes in the ventricles throughout gestation.

It is important to acknowledge that the current framework performs binary classification (normal vs. VM) and does not distinguish between isolated VM and cases associated with additional central nervous system anomalies. In clinical practice, differentiating isolated findings from complex malformations carries important prognostic and counseling implications. The present study was designed to first evaluate the feasibility and performance of an ultrasound-specific foundation model for VM detection. Future work may extend this approach toward multi-class diagnostic models that incorporate additional anatomical features and enable more granular stratification of fetal brain abnormalities.

In addition, semi-supervised or continual learning approaches can also be leveraged, taking advantage of the self-supervised nature of USF-MAE. As large amounts of unlabeled prenatal ultrasound data are stored in hospital archives, future versions of the model could be incrementally updated in the future to improve generalization without the need for manual annotations, while maintaining patient data privacy. On the clinical front, explainable artificial intelligence visualizations such as Eigen-CAM could be embedded into reporting interfaces to provide real-time decision support, highlighting which regions of the ventricles contribute to the model's classification and improving interpretability for sonographers and fetal medicine specialists.

Conclusion

Our study proposed a DL framework utilizing the USF-MAE foundation model for the automated classification of VM in prenatal ultrasound images. The model employed ultrasound-specific representations learned from large-scale, self-supervised pre-training and was subsequently fine-tuned on the fetal brain data set. The performance of our framework was evaluated using fivefold cross-validation and on independent test cohorts, and our results showed that our proposed approach achieved higher accuracy, precision, F_1 -score and specificity in classifying VM in prenatal ultrasound images compared with the baseline CNN and transformer models.

We can conclude that domain-specific pre-training on ultrasound data is beneficial for improving the generalization of features and label efficiency for downstream diagnostic tasks. The interpretability of the model was confirmed by Eigen-CAM heatmap visualizations, which revealed that the model's predictions were made based on relevant ventricular regions. These findings suggest that foundation models like USF-MAE can play an important role in enabling consistent and objective assessment of fetal brain structures on ultrasound, which may help improve the diagnostic reliability of prenatal screening for VM and other related neurological conditions with further validation on multi-institutional and multi-system data sets.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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